

HOSPITAL EMERGENCY ROOM ANALYSIS (Excel)

1) Problem Statement

We need to create a Hospital Emergency Room Analysis Dashboard in Excel to improve efficiency and provide useful insights . This dashboard will help stakeholders monitor , analyze, and make better decisions for managing patients and improving services .

Project Steps

1. Business Requirement Gathering
2. Understanding of Data
3. Data Connection (Import Data using Power Query)
4. Data Cleaning and Data Quality Check using Power Query
5. Creating Calendar Table using Power Query
6. Data Modeling using Power Pivot
7. Adding Required Columns (Calculated Columns using DAX in Power Pivot)
8. Creating PivotTables and Designing Dashboard Layout
9. Charts Development and Formatting
10. Dashboard / Report Development
11. Insights Generation

2) Dataset Description

This dataset contains detailed records of hospital patients, focusing on their admission details, demographic information, department referrals, and overall experience during their visit. It is useful for analyzing hospital operations, patient satisfaction, and waiting time efficiency.

Overview

Each row in the dataset represents a **single patient record**, capturing both **personal details** and **hospital service-related information**.

Column Name	Description
Patient Id	Unique identifier assigned to each patient for tracking and record management.
Patient Admission Date	The date and time when the patient was admitted to the healthcare facility.
Patient First Initial	The first initial of the patient’s first name (used for partial anonymization).
Patient Last Name	The surname of the patient.
Patient Gender	Gender of the patient (e.g., Male, Female).
Patient Age	Age of the patient at the time of admission.
Patient Race	Ethnicity or racial background of the patient (e.g., White, Asian, African American, etc.).
Department Referral	The medical department to which the patient was referred (e.g., General Practice, Orthopedics, Neurology).

Patient Admission Flag	Indicates whether the patient was officially admitted (TRUE/FALSE).
Patient Satisfaction Score	Rating given by the patient based on their experience (typically numerical scale).
Patient Waittime	Time (in minutes) the patient waited before receiving medical attention.
(Duplicate) Patient Admission Flag	Appears to be a repeated column; likely redundant and should be verified or removed during data cleaning.

3) Requirements

KPI's Requirement

1. Number of Patients:

- Count the total number of patients visiting the ER each day.
- Show a daily trend with an area sparkline to spot patterns like busy days or seasonal trends.

2. Average Wait Time:

- Find the average time patients wait to see a medical professional.
- Use an area sparkline to track daily changes and highlight days with longer wait times that might need improvements.

3. Patient Satisfaction Score:

- Check the average daily satisfaction score of patients to assess service quality.
- Use an area sparkline to show trends, spot drops in satisfaction, and link them to challenges.

Charts to Create:

1. Patient Admission Status: Show how many patients were admitted vs. not admitted.
2. Patient Age Distribution: Group patients by age.
3. Timeliness: Measure the percentage of patients seen within 30 minutes.
4. Gender Analysis: Display the number of patients by gender.
5. Department Referrals: Check which departments patients are referred to the most.

4) Data Cleaning & Analysis in power query

Step 1 : Understanding the data through excel

Step 2 : imported csv file to excel using power query

Step 3: transform data using power query

In power query editor –

- Click view to check the data quality
 - click on column quality
 - checked data type , empty in coulms

- (merge column) merged first name initial and surname by adding separator “. “
- (Replace Values) replaced values of F and M by Female and Male in Gender column , advanced options , match entire cell contents (helps in when you want to swap out only cells that exactly equal your search term — not partial matches.)
- for the Patient Admission Flag column , convert the Boolean type into text initially so that we can replace values TRUE as Admitted and FALSE as Not Admitted .
- removed repeated Patient Admission Flag column
- split the Patient Admission Date by delimmter (home – split column – by delimiter – replace comma by space – each occurrence of the delimiter)
- Converted the ‘Patient Admission Date’ column to Date format using the ‘Change Type with Locale’ option to handle inconsistent date formats correctly.
- renamed the columns

5) Creating calendar table using power query

- found the earliest and latest date (transform – date – earliest , latest) to understand the timeline of the data .

- to design a 2 years of calendar with unique dates

- we need calendar table as there are missing values in the column and repeated records

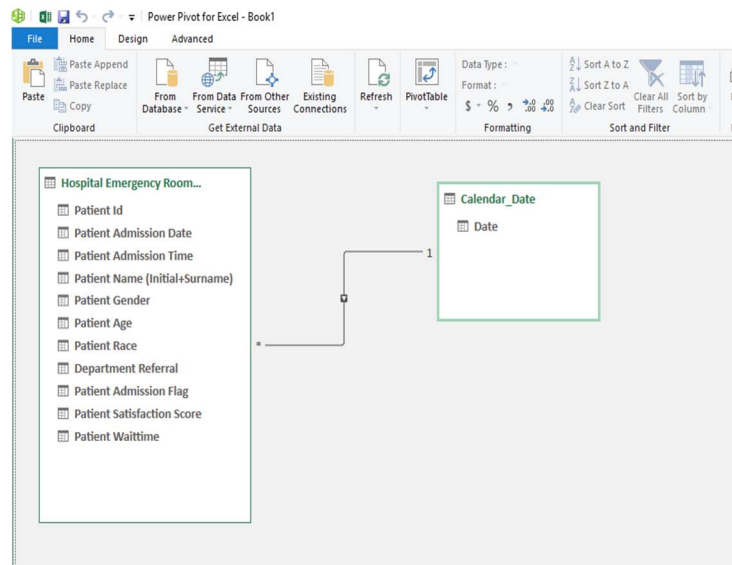
Steps to create :

- Home – new source – other sources – blank query
- Write the formula in blank query
- =List.Dates(#date(2023,01,01),731,#duration(1,0,0,0))
- This formula creates a list of daily dates starting from 1 January 2023 for the next 731 days, increasing one day at a time.
- the first value (1) represents 1 day, while the remaining zeros mean 0 hours, 0 minutes, and 0 seconds.
- Then once we get the list convert it into the table (to table) renamed the column to Date and table as calendar table .

6) Data Modelling using power pivot

- We have two tables Hospital Emergency and Calendar table and now we need to establish relation between two tables through common column date
- Closed and load to – tick the check box : only create connection (cause we don’t wanna load data) , add this data to data_modal (to use the data in power pivot we need to check this option)

- Diagram view (Created a relationship between the Calendar table and the main dataset using the Date column to enable time-based analysis. This helps in analyzing data by day, month, and year easily.)



- The one for the calendar tables indicates unique dates in it while the many symbol for the other table indicates duplicate dates in the column
- There is one to many cardinality/relationship

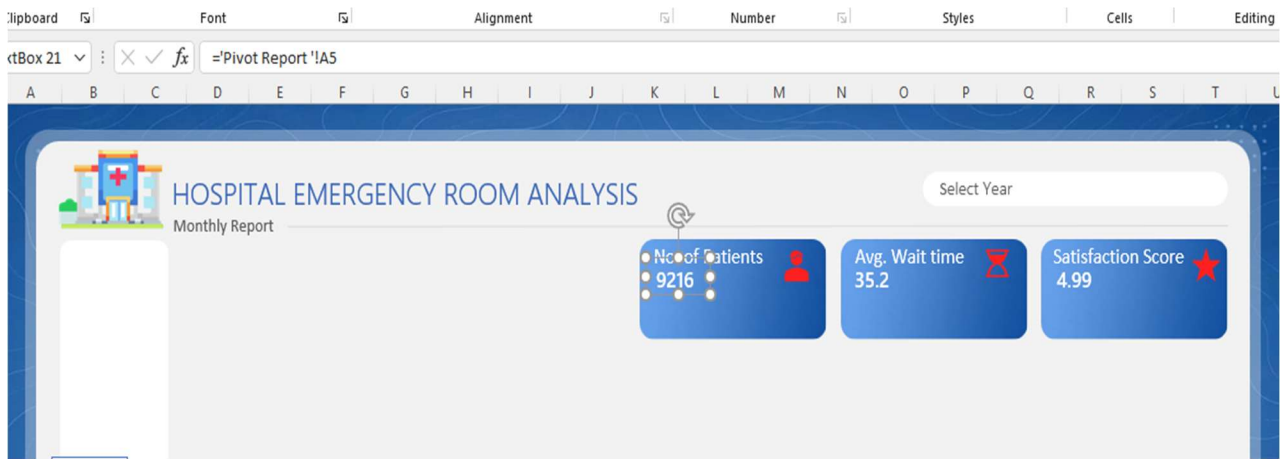
7) Adding calculated columns

1) Age Group Column

[Calculated Column] X ✓ fx =IF([Patient Age]>70,"70-79",IF([Patient Age]>60,"60-69",IF([Patient Age]>50,"50-59",IF([Patient Age]>40,"40-49",IF([Patient Age]>30,"30-39",IF([Patient Age]>20,"20-29",IF([Patient Age]>10,"10-19","0-09"))))))))									
	Patient Gender	Patient Age	Patient Race	Department Referral	Patient Admission Flag	Patient Satisfaction Score	Patient Waittime	Calculated Column 1	Add Column
1	Male	69	White	None	admitted		49	60-69	
2	Male	72	White	None	admitted		33	70-79	
3	Male	19	White	None	admitted		26	10-19	
4	Male	20	White	None	admitted		56	10-19	
5	Male	49	White	None	admitted		38	40-49	
6	Male	68	White	None	admitted		44	60-69	
7	Male	51	White	None	admitted		33	50-59	
8	Male	29	White	None	admitted		32	20-29	
9	Male	23	White	None	admitted		44	20-29	
10	Male	76	White	None	admitted		53	70-79	
11	Male	76	White	None	admitted		49	70-79	
12	Male	15	White	None	admitted		41	10-19	
13	Male	56	White	None	admitted		42	50-59	
14	Male	7	White	None	admitted		12	0-09	
15	Male	79	White	None	admitted		25	70-79	
16	Male	10	White	None	admitted		57	0-09	
17	Male	7	White	None	admitted		56	0-09	

- Created a calculated column to group patient ages into defined ranges (e.g., 0–9, 10–19, 20–29, etc.) using nested IF conditions to simplify age-based analysis.

2) Patient Attend Status



- Created a pivot table for month slicer
- Created a PivotTable with a month slicer using the Calendar table to analyze daily patient counts and enable interactive filtering by selected months.

The screenshot shows the 'PivotTable Fields' task pane on the left, which is configured with 'Date (Day)' in the Rows area and 'Count of Patient Id' in the Values area. The main table displays the following data:

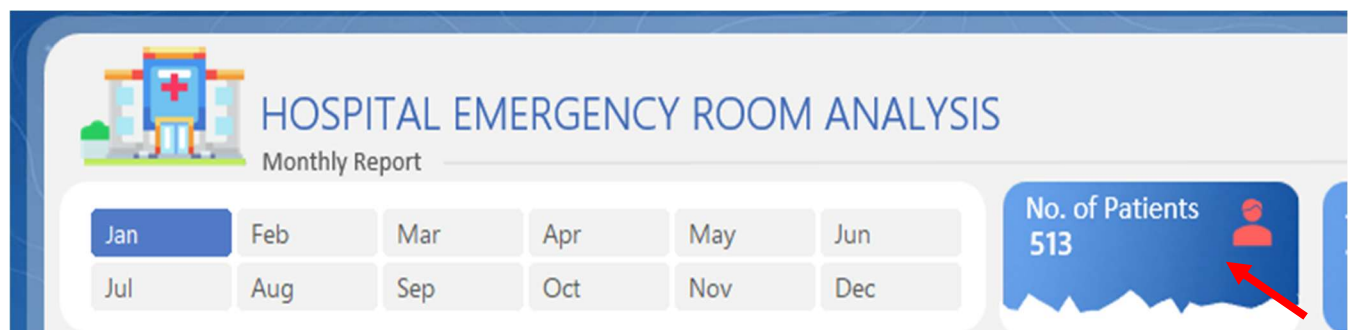
	No. of Patients	Average of Patient Waittime	Average of Patient Satisfaction Score
Count of Patient Id	9216	35.26	4.99

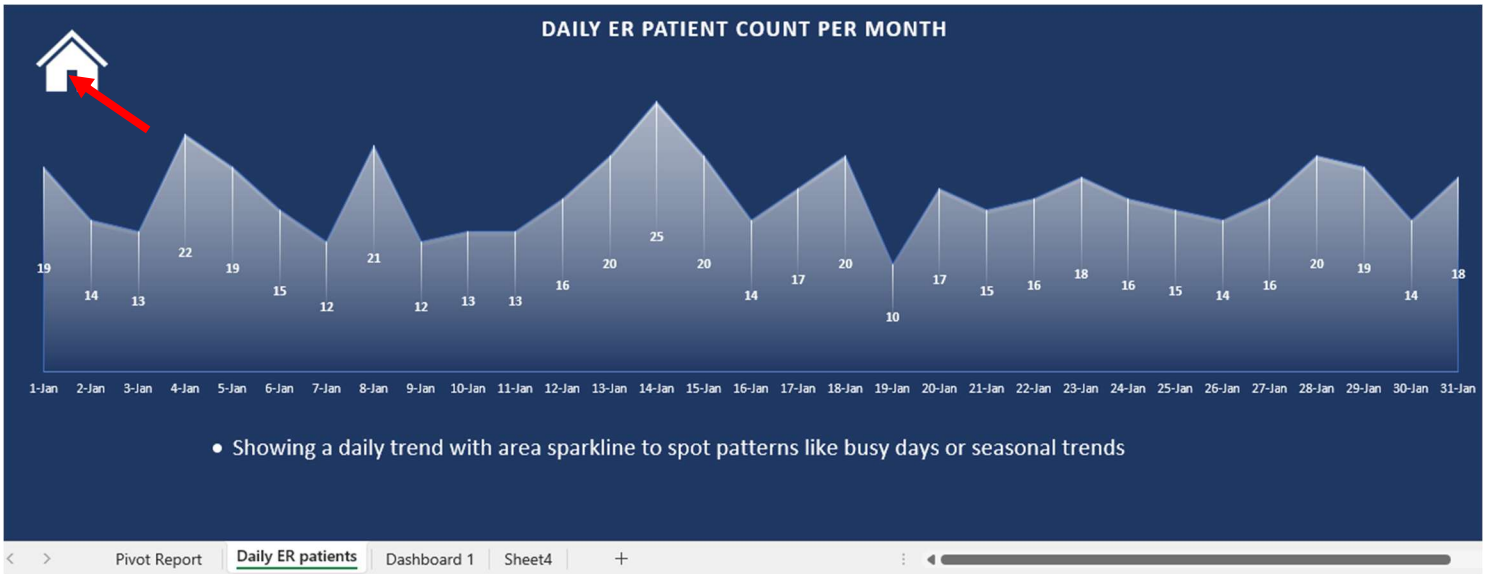
Below the PivotTable, a 'Date (Month)' slicer is shown with buttons for Jan, Feb, Mar, Apr, May, Jun, Jul, and Aug. To the right of the slicer, a table shows the daily patient counts for the first 18 days of January:

Row Labels	Count of Patient Id
1-Jan	19
2-Jan	14
3-Jan	13
4-Jan	22
5-Jan	19
6-Jan	15
7-Jan	12
8-Jan	21
9-Jan	12
10-Jan	13
11-Jan	13
12-Jan	16
13-Jan	20
14-Jan	25
15-Jan	20
16-Jan	14
17-Jan	17
18-Jan	20

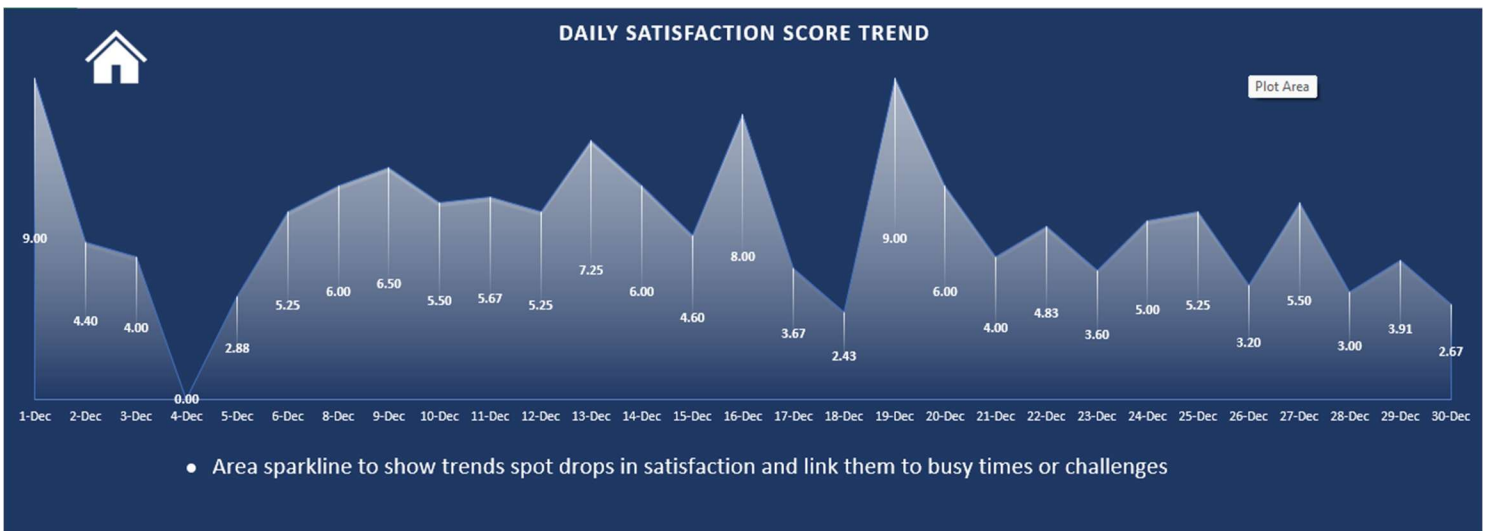
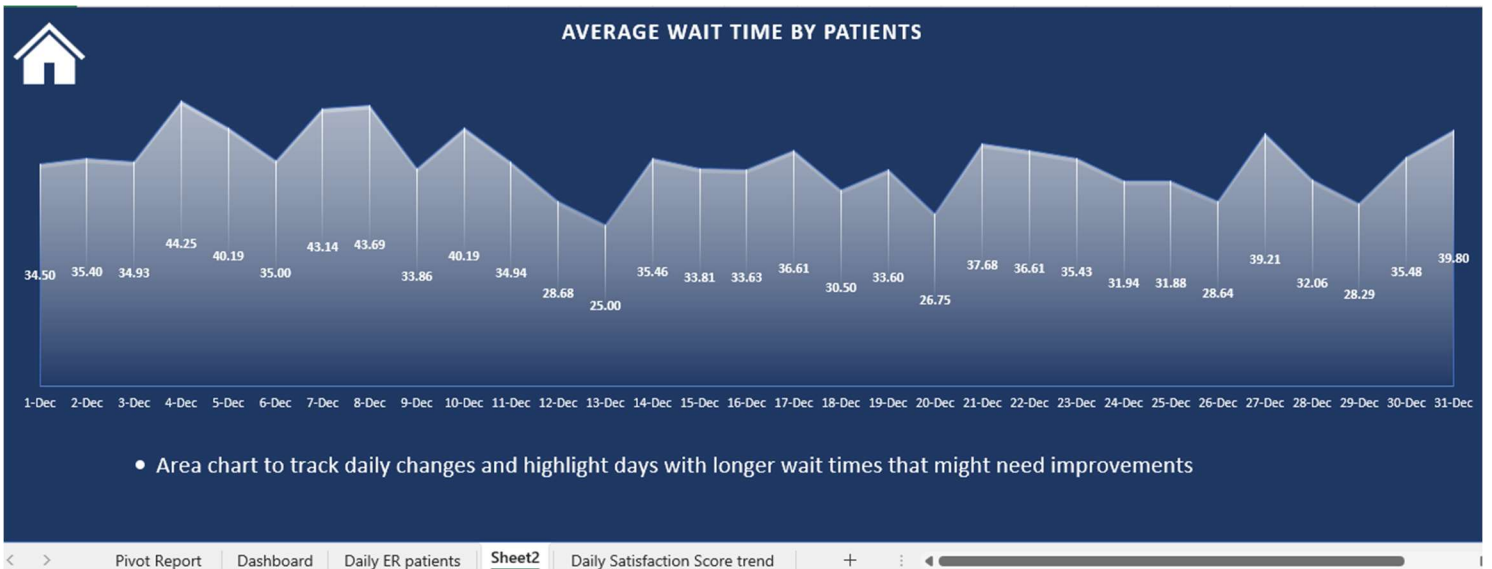
- Slicer – report connections – tick all the KPI pivots
- Connected the slicer to multiple PivotTables using the Report Connections option to enable synchronized filtering across all KPIs.
- Created a pivot chart for daily trend analysis of number of patients visited

Added hyperlink navigation to the KPI chart using Ctrl + K, enabling users to click the chart and directly navigate to the detailed 'Daily ER Patients' sheet for deeper analysis





Similarly for other two kpi s



- Created a PivotTable to analyze patient admission status by calculating the count and percentage of admitted and not admitted patients, providing a clear view of admission distribution.

Row Labels	Count of Patient Admission Flag	Count of Patient Admission Flag2
admitted	231	49.78%
not admitted	233	50.22%
Grand Total	464	100.00%

- Formatted the chart data labels to display percentage values using 'Value From Cells', ensuring clear representation of admission status distribution.
- Ensured dashboard consistency by connecting the slicer to all PivotTables so that all visuals update dynamically with user selection.
- Created multiple charts bar and column chart .

10) Dashboard , Insights and Recommendation



What makes it “Emergency Room Analysis”

Your data includes:

- **Patient Wait Time** → Critical in ER (urgent care)
- **Admission Status (Admitted / Not Admitted)** → ER decides who needs further treatment
- **Timeliness (Delay vs Ontime)** → Very important in emergency care
- **Department Referral** → ER redirects patients to departments
- **Patient Volume (daily/monthly)** → ER load analysis

All these are **core ER functions**

The Emergency Room is the **first point of contact** in a hospital for urgent cases.

So your dashboard is analyzing:

- How many patients come
- How fast they are treated
- Whether they are admitted or discharged
- How efficiently the ER is working

Insights :

- Almost each month in year 2023 – 24 patients load was moderate except for the august month in the year 2023 and the June month in the year 2024
- The approx. average waiting time over the months by the patients was 35 mins , the patients are facing delays not much but just a few minutes , but it adds to the time for doctors (Rec - Improve staff allocation during peak hours)
- Patients are facing more delays so (Rec - Hospital is not meeting service time standards)
- ~50% admitted / 50% not admitted , ER is handling both critical & non-critical cases equally (Improve triage system)
- For departmental referrals “None” is highest , Most patients are not referred → general treatment

- High wait time + high delay percentage
= **Operational inefficiency**

Recommendations :

1. Reduce Patient Wait Time

- Increase staff during peak hours
- Introduce fast-track system for minor cases

2. Improve Timeliness (Delay Issue)

- Set SLA (Service Level Agreement) for max wait time
- Monitor daily delay trends

3. Improve Patient Satisfaction Score

- Reduce wait time (biggest driver)
- Improve communication & service experience

The Hospital Emergency Room Analysis project provided valuable insights into patient flow, wait times, and overall service efficiency. The analysis revealed that the average patient wait time exceeds the expected threshold, indicating delays in service delivery. A significant proportion of patients are not attended within the desired time, highlighting operational inefficiencies.

Additionally, the majority of patients fall within the 20–40 age group, and a large number of cases are not referred to specific departments, suggesting either general treatment or gaps in data recording. The patient admission rate shows a balanced distribution, indicating a mix of critical and non-critical cases handled by the emergency room.

Overall, the dashboard helps in understanding key performance areas and identifying opportunities for improving emergency room operations.

Recommendations

- **Reduce Wait Time:** Increase staff availability during peak hours and introduce a fast-track system for less critical patients.
- **Improve Timeliness:** Set service-level benchmarks and continuously monitor delay metrics to ensure patients are attended within acceptable time limits.
- **Optimize Resource Allocation:** Analyze peak patient inflow periods and allocate medical staff accordingly to handle demand efficiently.
- **Enhance Referral Process:** Standardize department referral procedures and reduce the number of “None” entries to improve data accuracy.
- **Focus on Patient Demographics:** Investigate the high number of patients in the 20–40 age group and implement preventive healthcare measures.
- **Improve Patient Satisfaction:** Reduce waiting time and enhance patient interaction to improve overall experience.
- **Ensure Data Quality:** Implement stricter data entry validation to minimize missing or inconsistent values.
- **Future Scope:** Develop advanced dashboards using Power BI and incorporate real-time data monitoring for better decision-making.